

Chapter 165 — Management of Recurrent and Newly Metastatic Prostate Cancer

IF YOU READ NOTHING ELSE

- **BCR (biochemical recurrence) after RP (radical prostatectomy):** AUA (American Urological Association) PSA ≥ 0.2 ; NCCN (National Comprehensive Cancer Network) ≥ 0.1 /non-undetectable/two rises.
- **BCR after EBRT (external beam radiotherapy):** Phoenix = nadir + ≥ 2 ng/mL; after LDR (low-dose-rate) brachytherapy $\rightarrow \geq 0.2$.
- **Salvage RT (radiotherapy) = only curative option** for post-RP BCR.
- **Early salvage at PSA 0.1–0.2 $\equiv$ adjuvant** (RADICALS/RAVES/GETUG-AFU 17); $\geq 60\%$ spared RT.
- **Salvage RT dose ≥ 64 –65 Gy fossa; SAKK 09/10: 64 = 70 Gy.**
- **RT0G 0534:** bed $\rightarrow$ bed + ADT (androgen deprivation therapy) $\rightarrow$ pelvis + ADT = 70.9 $\rightarrow$ 81.3 $\rightarrow$ 87.4% 5-year FFP (freedom from progression).
- **Imaging before salvage:** PET (positron emission tomography)/CT excludes metastases; mpMRI (multiparametric MRI) finds local.
- **Castrate testosterone < 50 ng/dL;** rising PSA with castrate T = CRPC (castration-resistant prostate cancer).
- **CHAARTED high volume:** ≥ 4 bone mets, ≥ 1 outside axial skeleton, or any visceral.
- **mHSPC (metastatic hormone-sensitive prostate cancer):** ADT alone insufficient $\rightarrow$ add docetaxel, abiraterone or ARTA (AR-targeting agent).

CURATIVE INTENT APPROACHES TO BCR — AFTER PROSTATECTOMY

- **Cut-offs:** BCR by 10 years ~25–40%; high-risk ~60% at 5 years.
- **Threshold most predictive of metastasis:** PSA ≥ 0.4 then rising.
- **EAU (European Association of Urology) low-risk BCR:** PSA-DT (doubling time) > 1 year AND grade group < 4 ; high-risk: PSA-DT ≤ 1 year OR GG 4–5.
- **Ultrasensitive PSA:** prognostic mainly in intermediate/high risk; ~half of early detectable values fall $\rightarrow$ watch, don't treat.
- **Only modifiable BCR risk factor:** positive margin; > 1 mm length, Gleason ≥ 4 at margin worse.
- **pT3a margin-negative** lower BCR risk than pT2 margin-positive.
- **Nomograms:** CAPRA pre-op; CAPRA-S post-op; MSK tools $\rightarrow$ 15-year CSS (cancer-specific survival).
- **Pound (1999):** BCR $\rightarrow$ mets 8 years; mets $\rightarrow$ death 5 years.
- **Pound predictors:** time to BCR < 2 years, PSA-DT < 10 months, Gleason 8–10.
- **Post-RP PSA follow-up lifelong** — $\sim 1/4$ recur after 5 years.
- **Bone scan/CT:** low yield $< \text{PSA } 10$.
- **Choline PET:** ~half positive $< \text{PSA } 1$; fluciclovine poor < 0.5 , ~nil < 0.2 .
- **PSMA $>$ fluciclovine** at PSA < 0.5 .
- **EMPIRE-1:** fluciclovine-guided salvage $\rightarrow$ 3-year control 75.5 vs 63.0%.
- **Salvage RT CSS benefit:** within 2 years of BCR, PSA-DT < 6 months.
- **Each 0.1 ng/mL pre-salvage $\rightarrow$ ~2.6% RFS lost.**
- **Dose:** > 66 Gy better in series; ~2%/Gy; nodes 40–45 Gy.
- **Margin > 3 mm in GG 3–5 $\rightarrow$ poorer salvage survival.**
- **No MRI lesion before salvage $\rightarrow$ higher failure** (occult mets).
- **RT0G 9601:** RT + 2 years bicalutamide $\rightarrow$ OS 76.3 vs 71.3% (12 years).
- **GETUG-AFU 16:** RT + 6 months goserelin $\rightarrow$ BCR/MFS better, OS not.
- **Second BCR after salvage RT:** ~13–38% $\rightarrow$ lifelong PSA.

AFTER DEFINITIVE RADIATION THERAPY

- **Phoenix (ASTRO/RT0G 2005):** nadir + ≥ 2 ; valid with ADT.
- **LDR-BT cure:** PSA < 0.2 at 4 years; BT dose $\geq 2 \times$ EBRT.
- **PSA bounce:** rise 0.1–0.5, ~9 months, then falls $\rightarrow$ favourable; monitor.
- **Progression predictors:** short PSA-DT, stage, Gleason, short interval to BCR.

- **Post-RT CT:** useful only PSA > 20 ; TRUS (transrectal ultrasound) $\approx$ DRE.
- **T2 MRI poor after RT** (diffuse low signal) $\rightarrow$ DCE + DWI (mpMRI), AUC ~0.84.
- **Biopsy mandatory** before local salvage; post-RT Gleason unreliable.

TREATMENT OF POSTRADIATION BCR

- **All salvage requires:** biopsy-proven local, imaging M0, life expectancy 5–10 years+.
- **EAU salvage RP criteria:** LE ≥ 10 years, PSA < 10 , GG ≤ 3 , N0, initial cT1–2.
- **Salvage RP:** robotic $\rightarrow$ better continence, fewer contractures/rectal injuries.
- **Salvage RP poor prognosis:** GG ≥ 3 , pre-salvage PSA ≥ 4 .
- **Salvage BT:** only after EBRT; LDR $\approx$ HDR (high-dose-rate); grade ≥ 3 GU ~17–24%.
- **Cryotherapy selection:** PSA < 10 (ideally < 4), PSA-DT ≥ 16 months.
- **Cryotherapy:** urethral warming $\rightarrow$ incontinence single digits; fistula 1–3.4%.
- **Cryo nadir < 0.2 $\rightarrow$ avoids hormones.**
- **HIFU (high-intensity focused ultrasound):** ≥ 2 years after EBRT; fistula, osteitis pubis.
- **All modalities ~50–70% biochemical control** (Table 165.2).
- **After salvage for RT failure:** recurrence $> 50\%$.
- **Focal therapy failure $\rightarrow$ manage as radiation failure.**

NONCURATIVE INTENT APPROACHES TO BCR

- **AUA: observation over ADT;** ADT if PSA-DT < 12 months or high grade; intermittent preferred.
- **Untreated post-RP BCR (Pound):** only ~1/3 metastasise.
- **EAU observation (post-RT):** interval to failure > 18 months, GG < 4 .
- **Observation imaging:** PET/CT preferred, esp. PSA-DT < 12 months.
- **AR-V7 (lacks LBD) $\rightarrow$ castration resistance, poor survival.**

METHODS OF ANDROGEN DEPRIVATION THERAPY

- **Orchidectomy:** testis = 95% T; castrate ~6 hours; < 20 ng/dL by 12 hours.
- **Orchidectomy:** fewer cardiac, PAD, fracture events than LHRH agonists $\rightarrow$ offer if lifelong ADT.
- **NSAA (nonsteroidal antiandrogen) alone:** T $\uparrow$ ~1.5 $\times$ $\rightarrow$ gynaecomastia (aromatization).

- **All NSAAs:** hepatotoxicity, pneumonitis (nilutamide 1–2%).
- **Flutamide:** diarrhoea, 250 mg tds; inferior OS to DES.
- **Nilutamide:** delayed dark adaptation.
- **Bicalutamide:** 50 mg + LHRH only approved; 150 mg → worse OS localised.
- **Enzalutamide:** AFFIRM 2012; seizures <1%; approved all CRPC, mHSPC, high-risk nmHSPC.
- **Apalutamide:** SPARTAN 2018, MFS 40.5 vs 16.2 (HR 0.28); rash, hypothyroidism, falls.
- **Darolutamide:** ARAMIS 2019, OS HR 0.68; mHSPC only with docetaxel.
- **nmCRPC ARTA indication:** PSA-DT ≤10 months.
- **AAWS (antiandrogen withdrawal syndrome):** 15–30% PSA fall >50%, 3.5–5 months; enzalutamide 0–10%; no OS gain.
- **LHRH agonist flare:** LH ↑ 10×, 10–20 days; antiandrogen cover 21–28 days.
- **Flare danger:** spinal/weight-bearing bone mets.
- **Agonist** ↑ FSH; **antagonist** ↓ FSH.
- **Antagonist:** LH ↓ 84% in 24 hours, no flare → cord compression.
- **Abarelix:** allergy → observe ≥30 minutes.
- **Degarelix:** castrate >95% day 3 vs 0% leuprolide.
- **HERO (relugolix, oral):** castrate day 4 56 vs 0%; T recovery 54 vs 2%.
- ~50% fail to recover T 5 years after LHRH agents.
- **Aminoglutethimide:** replace cortisone + fludrocortisone.
- **Ketoconazole:** castrate <4 hours, 200–400 mg 8-hourly + hydrocortisone; emergency use.
- **Abiraterone:** CYP17 (17α-hydroxylase, 17,20-lyase) + prednisone 5 mg → prevents ↑ ACTH → ↓ K⁺, ↑ BP, oedema.
- **COU-AA-301** OS 14.8 vs 10.9; **COU-AA-302** rPFS 16.5 vs 8.3.

TIMING OF THERAPY

- **Primary ADT for localised disease:** not recommended.
- **Messing:** pN+ post-RP → immediate ADT OS benefit (bulky, dated).
- **EORTC 30891:** 30% delayed never needed ADT; benefit if PSA >50 or PSA-DT <12 months.
- **TOAD:** immediate ADT OS HR 0.55.
- **EMBARC entry:** PSA-DT ≤9 months; PSA ≥1 post-RP or ≥2 above nadir post-RT.
- **EMBARC MFS:** combo 87.3% > enza 80.0% > leuprolide 71.4% (HR 0.42).
- **EMBARC pause/restart:** stop if PSA <0.2 at week 37; restart 2 (post-RP) / 5 (post-RT).
- **Intermittent after RT (NCIC PR.7):** non-inferior; stop <4, restart 10.
- **Intermittent in M1 (SWOG 9346):** inconclusive → may be worse; restart 20.
- **Calcium 1000–1200 mg + vitamin D₃ 400–1000 IU** for all on ADT.
- **Treat bone if:** T-score –1 to –2.5 + 10-year hip >3% or major >20%.
- **Denosumab 60 mg SC 6-monthly;** zoledronate 4–5 mg IV yearly; alendronate 70 mg weekly.
- **Hot flushes 50–80%;** oestrogens most effective (~90%); venlafaxine taper.
- **Libido worse than erection;** PDE5 inhibitors ineffective.
- **ADT metabolic syndrome:** subcutaneous fat, ↑ HDL; diabetes HR 1.44 (agonist).

- **Gynaecomastia:** DES/bicalutamide 150 ~50%; orchidectomy/agonist ≤5%.
- **Breast RT prophylaxis 10–15 Gy single** — useless once established.
- **Anaemia:** normochromic normocytic; Hb <12 pre-ADT → worse survival; recovery up to 1 year.

FOLLOW-UP ON ADT (NONMETASTATIC)

- **Review 3–6-monthly:** FBC, chemistry, PSA.
- **Faster time to PSA <0.2** → better survival.
- **HSD3B1 1245A>C (15–35%)** → ADT resistance; homozygous ~2× mets.

HORMONE SENSITIVE METASTATIC PROSTATE CANCER

- **Genetic testing all M1:** >10% DNA-repair mutations.
- **Median OS:** historical ~30 months → now >48 months.
- **SWOG 9346 nadir at 7 months:** ≤0.2 → 75; 0.2–4 → 44; >4 → 13 months.
- **Observation M1:** only if asymptomatic, LE <5 years.
- **CAB with NSAA:** AUA against.
- **Docetaxel 75 mg/m² 3-weekly × 6.**
- **CHAARTED:** OS HR 0.61, high volume only; GETUG-AFU 15 negative.
- **Docetaxel meta-analysis:** HR 0.77, 94 fewer deaths/1000.
- **LATITUDE high risk (≥2):** Gleason ≥8, ≥3 bone lesions, visceral; OS HR 0.62.
- **STAMPEDE G:** abiraterone OS HR 0.63.
- **ARCHES** rPFS 0.39; **ENZAMET** OS 0.67 (vs NSAA); **TITAN** OS 0.67.
- **ARTAs/abiraterone benefit regardless of volume.**
- **PEACE-1:** abiraterone + docetaxel → OS HR 0.75.
- **ARASENS:** darolutamide + docetaxel → OS HR 0.68.
- **STAMPEDE H:** prostate RT → OS HR 0.68 low volume only.
- **Oligometastatic = ≤3 sites on PET.**
- **STOMP:** ADT-free 21 vs 13 months; **ORIOLE:** 6-month progression 19 vs 61%.
- **mHSPC follow-up 3–6-monthly;** T with every PSA rise.

CLASSIC TRAPS

- **Trap:** detectable PSA after RP = metastatic course — **Actually:** ~half never fail clinically.
- **Trap:** Phoenix applies to brachytherapy — **Actually:** after LDR-BT use ≥0.2.
- **Trap:** post-RT PSA bounce = failure — **Actually:** bounce predicts better outcome.
- **Trap:** NSAA alone produces castration — **Actually:** testosterone rises ~1.5×.
- **Trap:** LHRH antagonist needs antiandrogen cover — **Actually:** no flare; agonists need 21–28 days.
- **Trap:** abiraterone prednisone replaces aldosterone — **Actually:** it suppresses ACTH.
- **Trap:** docetaxel helps all mHSPC — **Actually:** CHAARTED benefit high volume only.
- **Trap:** prostate RT for all M1 — **Actually:** low-volume only (STAMPEDE H).
- **Trap:** rising PSA on ADT = CRPC — **Actually:** check testosterone <50 ng/dL first.
- **Trap:** SPPORT (RTOG 0534) showed OS benefit — **Actually:** endpoint freedom from progression.